

ICT/18/6859

The Federal University of Technology, Akure
School of Engineering and Engineering Technology
Department of Information and Communication Technology
Second Semester Examinations 2023/2024 Session
ICT 504, Artificial Neural Network (ANN): [3 Units]
Date: 25th of September, 2024: 3 HOURS.



General Instruction: Answer ALL QUESTIONS in Section A and ONE (1)
QUESTION EACH from Section B, C and D.

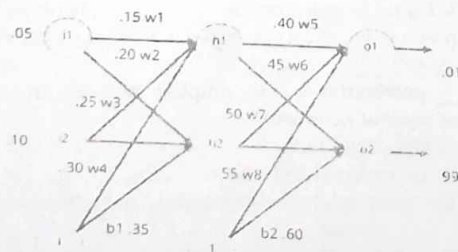
SECTION A

- 1) What is the basic building block of an Artificial Neural Network (ANN)? (A) Perceptron (B) Neuron (C) Synapse (D) Link.
- 2) What is the main function of an activation function in a neural network? (A) Normalize data (B) Introduce non-linearity (C) Initialize weights (D) Adjust learning rate
- 3) Which of the following is an example of a feedforward neural network? (A) Perceptron (B) Hopfield Network (C) Recurrent Neural Network (D) Kohonen Network
- 4) The process of adjusting the weights in an artificial neural network is known as: (A) Activation (B) Propagation (C) Learning (D) Inference
- 5) Which of the following represents the input layer in a neural network? (A) Hidden layer (B) Output layer (C) Source nodes (D) Synaptic weight
- 6) What is a major advantage of artificial neural networks? (A) They require no training data (B) They can model complex relationships (C) They have very simple architecture (D) They do not overfit
- 7) The term "learning rate" in neural networks refers to: (A) How quickly the network learns (B) The size of weight updates (C) The speed of processing (D) Number of layers in the network
- 8) In the context of neural networks, overfitting refers to: (A) Underestimating data complexity (B) Failing to converge (C) Fitting too closely to training data (D) Ignoring training data
- 9) What is the purpose of the bias in a neuron? (A) Control the number of inputs (B) Shift the activation function (C) Reduce overfitting (D) Improve data scaling
- 10) What kind of data can neural networks process? (A) Only numerical data (B) Only categorical data (C) Both numerical and categorical data (D) Only image data
- 11) In an ANN, what type of learning involves labeled data?.....
- 12) A neural network learns by adjusting the _____ between the neurons.
- 13) The process of propagating the error backwards through the network to update the weights is known as Backward
- 14) The _____ perceptron is the simplest type of artificial neural network, consisting of only one layer of neurons.
- 15) What does ADALINE stand for?
- 16) For weight updating in ADALINE, the _____ rule is used
- 17) In ADALINE, the error term is computed as the difference between the actual output and the _____ output.
- 18) The main objective of the ADALINE model is to minimize the _____, which represents the squared difference between the predicted output and the actual output.
- 19) The learning algorithm used to minimize the loss function in a single-layer linear regression model is called _____

- 20) Ultimate results are produced in ANN at the _____ layer?
- 21) Which type of learning process in ANN learns with a teacher _____
- 22) Which type of learning process in ANN learns without a teacher _____
- 23) The layers between the input and output layers in a multi-layer neural network are called _____ layers.
- 24) The process of feeding the output of one layer as the input to the next layer in a neural network is known as _____ propagation
- 25) Which activation functions is commonly used in the output layer of a multi-layer ANN for binary classification tasks

SECTION B

1. (a). With diagrams, explain the basic architecture of the McCulloch-Pitts (M-P) neuron model. How does it process input signals to produce an output. (7marks)
- (b). Why is it important to choose an appropriate learning rate in the perceptron learning algorithm? What happens if the learning rate is too high or too low. (5 marks)
- (c). Train a Single-layer perceptron NN using the logic gates AND truth table with binary ON/OFF as (1, -1). The initial weights and bias are zero. The learning rate is 1. The activation function is a signum function (1, 0, -1). Stop at Epoch 2 and state the inputs which are classified correctly and those classified incorrectly. Draw the SLP-NN model for the AND logic gate indicating the modified weights and bias. What is your conclusion based on your result. (13 marks)
2. (a) (i) Define the term "Neural Network" (1 mark)
- (ii). List and briefly discuss 4 useful properties of Neural Networks (4 marks)
- (b). Briefly discuss why the perceptron model cannot solve the XOR problem. (5 marks)
- (c). For the MLP-NN given below, using the backpropagation algorithm, update w_1 and w_5 for one iteration. The initial parameters are given as follows: **Inputs:** $x_1 = 0.05$, $x_2 = 0.10$. **Initial weights:** $w_1 = 0.15$, $w_2 = 0.20$, $w_3 = 0.25$, $w_4 = 0.30$, $w_5 = 0.40$, $w_6 = 0.45$, $w_7 = 0.50$, $w_8 = 0.55$. **Initial biases:** $b_1 = 0.35$, $b_2 = 0.60$. Target values: $T_1 = 0.01$, $T_2 = 0.99$, learning rate = 0.5. Activation function is sigmoid function. (15 marks)



low error
high learning

the value of w_1 and w_5 are 0.15 and 0.40 respectively.

and 1

low

SECTION C

3. (a). Explain the difference between a single-layer perceptron and a multi-layer perceptron (4 marks)
- (b). What is backpropagation, and how does it work in a multi-layer neural network. (6 marks)
- (c). A school wants to predict their students final exam grades based on the number hours they studied. The dataset is given in the table below. Given the following initial parameters: weight $w = 0$, bias $b = 0$, learning rate $= 0.01$, cost function is MSE. Use a single-layer linear regression to obtain the line of best fit. Implement the Gradient Descent Algorithm to adjust the weights and bias. Solve for 2 iterations. Draw the old and new line of best fit and comment on your results. (15 marks)

Hours of Study	Scores
3	40
5	55
8	65
11	80

4. (a). Briefly discuss the limitations of the McCulloch-Pitts neuron model. What are some key features that it lacks compared to more modern neural network models. (7 marks)
- (b). What is the key difference between the ADALINE model and the Perceptron model in terms of the learning process. (3 marks)
- (c). Use the ADALINE model to solve the question in (3c). The initial weight is 1, initial bias is 35, learning rate is 0.01, error function is MSE, stopping error condition is 85, stopping epoch is 2. State your final weight and bias obtained, draw the ADALINE model corresponding to your new weight and bias. Draw the old and the new line of best fit on the same graph and comment on your observation after the training. (15 marks)

SECTION D

5. (a). Describe the role of thresholding in the McCulloch-Pitts neuron model. Why is it necessary, and how does it influence the output of the neuron? (5 marks)
- (b). List the three fundamental network architectures of neural networks, state one notable feature about each, and draw the block diagram for each of them. (9 marks)
- (c). You are tasked with classifying an email as either "spam" or "not spam" using a M-P NN model. The input features are: Keyword "Free" in Subject (1 if present, 0 if not), "More than 3 links in the email" (1 if true, 0 if false), and "Sender is unknown" (1 if sender is not in the contact list, 0 if they are). The given inputs for three different emails are as follows:

010100 (000000)

3

Single → No bias layer

multiple hidden

Output hidden, one to input

Email	Keyword "Free"	More than 3 links	Sender is unknown
Email 1	1	1	1
Email 2	0	1	1
Email 3	1	0	0

You are provided the weights for each input feature as follows: Weight for Keyword "Free" = 0.5, weight for "More than 3 links" = 0.7, Weight for "Sender is unknown" = 0.6.

The perceptron has a threshold of $0 \geq 1.0$, and the activation function is a binary step function such that output 1 is a "spam" if the weighted sum is greater than or equal to the threshold and output 0 "not spam" if the weighted sum is less than the threshold.

- Calculate the weighted sum for each email. (6 marks)
- Draw the M-P NN model for this task (2 marks)
- Determine whether each email is classified as "spam" or "not spam." (3 marks)

6. (a). Why is the bias term important in the perceptron model, and how does it influence the decision boundary? (4 marks)

(b). List the three learning process in ANN. Mention one distinct feature about each and state the sub-categories of tasks carried out under each of them. (9 marks)

(c). Using the output data obtained in question (5c) as the target value, train the dataset using a Single-layer perceptron neural network model with a learning rate = 1, and bias = 0.4. Draw the SLP-NN model with the optimal weights obtained. Test the following email inputs with your SLP-NN and classify them appropriately: (0,0,1), (1,0,1) and (0,0,0). (12 marks)

Email	Keyword "Free"	More than 3 links	Sender is unknown	Target value
Email 1	1	1	1	1
Email 2	0	1	1	1
Email 3	1	0	0	0



THE FEDERAL UNIVERSITY OF TECHNOLOGY, AKURE
SCHOOL OF ELECTRICAL SYSTEMS ENGINEERING
DEPARTMENT OF INFORMATION AND COMMUNICATION ENGINEERING

B. ENG. DEGREE EXAMINATIONS. SECOND SEMESTER 2024/2025 SESSION
COURSE CODE: ICT 504 COURSE TITLE: Smart Intelligence Systems -
Machine Learning

UNITS: 3

TIME ALLOWED: 3 Hours

Instruction: Answer ALL QUESTIONS in Section A and ONE QUESTION EACH from Section B, C and D. All TABLES are shown in Page 4.

SECTION A

1. In simple linear regression, the slope coefficient represents: (A) Change in x for one unit change in y (B) Change in y for one unit change in x (C) The intercept term (D) The error variance.
2. Which algorithm minimizes the log-loss (cross-entropy) function? (A) Linear Regression. (B) KNN Regression (C) Logistic Regression. (D) Naïve Bayes
3. In batch gradient descent, gradients are computed using: (A) One randomly chosen instance (B) A mini subset of instances (C) The entire dataset (D) No dataset at all
4. In stochastic gradient descent, parameter updates occur: (A) After each epoch (B) After each batch (C) After each sample (D) Only once
5. In KNN classification, prediction is made by: (A) Computing weighted averages (B) Majority vote of nearest neighbors (C) Fitting a straight line (D) Minimizing cost function
6. The coefficient of determination (R^2) measures: (A) Proportion of variance explained by the model (B) Correlation between errors and inputs (C) Probability of classification error (D) Bias in regression
7. If the learning rate in gradient descent is too large, the algorithm may: (A) Converge smoothly (B) Never update weights (C) Stop immediately (D) Overshoot and diverge
8. Logistic regression outputs: (A) Class labels directly (B) Probabilities of class membership (C) Cluster assignments (D) Decision trees
9. The gradient of the cost function points in the direction of: (A) Maximum increase (B) Maximum decrease (C) Zero slope (D) Random movement
10. Increasing the number of neighbors K in KNN generally makes the decision boundary: (A) More flexible (B) More complex (C) Smoother (D) Less interpretable
11. The three main types of machine learning are supervised learning, unsupervised learning, and reinforcement learning.
12. In supervised learning, the algorithm is trained on input-output pairs called training data.
13. A model that predicts continuous values is called a regression model.
14. Logistic regression is primarily used for classification problems.
15. In linear regression, the best-fit line is obtained by minimizing the sum of squares of errors
16. In KNN classification, the predicted label is chosen by majority vote among neighbors.
17. The threshold commonly used for classification in logistic regression is 0.5.
18. Batch gradient descent computes gradients using the entire dataset at each update.
19. Stochastic gradient descent computes gradients using one sample(s) at each update.
20. The learning rate (α) controls the size of the parameter updates in gradient descent.
21. Linear regression is suitable for predicting categorical outcomes like "Pass" or "Fail." (True/False)
22. In supervised learning, the algorithm learns from labeled examples of input-output pairs. (True/False)
23. Batch gradient descent updates parameters after processing each training sample. (True/False)
24. Logistic regression directly outputs class labels (0 or 1) without using probabilities. (True/False)
25. KNN classification requires calculating distances between the query point and all training samples. (True/False)

SECTION B

1. a. i). In one sentence, state the difference between a *traditional computer program* and a *machine learning program*. (1 mark)
 - ii). What do you understand by the term *model-based learning strategy* in machine learning. (2 marks)
 - iii). List the five main types of machine learning and provide two examples of each. (5 marks)
 - b. i). With the aid of equations, explain how logistic regression predicts probabilities. (3 marks)
 - ii). What does the 'K' represent in K-Nearest Neighbors (KNN) and what is the role of a distance metric in KNN classification. (2 marks)
 - iii). State one advantage of KNN in classification tasks. (1 mark)
 - c). A university lecturer collected data on students' *library visits per week* (X_1) and *assignment submission rate (%)* (X_2), alongside their GPA category (Y), where $High \geq 3.5$ and $Low < 3.5$ as shown in *Table 1*. A new student has $(X_1, X_2) = (3, 72)$
 - i). Classify Student K using $K = 3$ (show all steps). (4 marks)
 - ii). Classify Student K using $K = 5$ (show all steps). (4 marks)
 - iii). Plot and compare your results in (i) and (ii), and explain why they differ. (3 marks)
 2. a. i). State one real-life application of supervised learning. (1 mark)
 - ii). Explain the term *gradient descent* in machine learning. (2 marks)
 - iii). In machine learning, what are *features*? Provide an example. (2 marks)
 - b. i). Derive the gradient descent update rule for linear regression. (4 marks)
 - ii). What is the main difference between KNN classification and KNN regression. (1 mark)
 - iii). What type of output variable is required in KNN regression. (1 mark)
- A university lecturer records *Study Hours/week* (X_1), *Lecture Attendance (%)* (X_2), and *Exam Score* (Y) out of 100 for 10 students as shown in *Table 2*. A new student K has $(X_1, X_2) = (5.5, 77)$. Use $K = 5$ neighbours and answer the following:
- c. i). Transform attendance as $X_2' = X_2/10$. Write the scaled query point (X_1, X_2') . Then compute Euclidean distances from K to all students in the scaled space and list the 5 nearest neighbors (show numbers to 2 d.p.). (5 marks)
 - ii). Using the 5 nearest neighbors identified in Q1, compute the unweighted prediction and the weighted prediction with weights $w_i = 1/d^2$. (5 marks)
 - iii). Compare the two predictions and briefly explain why they differ. (4 marks)

SECTION C

3. a. i). Define the term "*overfitting*" in machine learning. (1 mark)
 - ii). State two advantages of using mini-batch gradient descent compared to batch or stochastic gradient descent. (2 marks)
 - iii). The main components of a typical ML model are *Task* (T), *Experience* (E), and *Performance* (P). Briefly explain what each of these terms means in machine learning. (3 marks)
 - b. i). List the two stopping criterion in gradient descent algorithm. (2 marks)
 - ii). Compare gradient descent with the closed-form solution for Simple linear regression. Under what conditions would you prefer gradient descent? (2 marks)
 - iii). Highlight ONE advantage and disadvantage of the three variants of the Gradient Descent Algorithm (3 marks)
- A university lecturer collects the following data on *study hours per week* X and *exam score* Y (out of 100) as shown in *Table 3*. Determine the SLR equation
- c. i). Determine the SLR equation using the *classical formula* approach. (4 marks)
 - ii). Obtain the SLR equation using the *ML Vectorized close-form equation* approach. (4 marks)
 - iii). Sketch the graphs and compare your results in (i) and (ii) and predict the scores a student will obtain if he studies for 5 hours. (4 marks)
4. a. i). What is the role of a *cost (loss) function* in machine learning. (2 mark)
 - ii). List the main steps in a typical machine learning project workflow. (3 marks)

- iii). List three ML-powered applications you can find on a typical smartphone. (1 mark)
- b). i). What is the meaning of *residual* in simple linear regression (SLR), and why is the least squares method used in estimating the regression line. (2 marks)
- ii). Differentiate between the roles of the independent variable (x) and the dependent variable (y) in simple linear regression. (2 marks)
- iii). What is the main objective of simple linear regression and what does β_1 represent in simple linear regression equation. (3 marks)
- A university lecturer wants to model the relationship between *study hours* (X) and *exam scores* (Y) using simple linear regression. The following dataset is collected and shown in **Table 4**. With initial parameters: $\theta_0 = 0$, $\theta_1 = 0$, Learning rate $\alpha = 0.1$, Epoch = 3.
- c). i). Use the *Batch Gradient Descent* (BGD) to determine the parameters θ_0 , θ_1 fitting the SLR equation. (5 marks)
- ii). Comment briefly on what is happening to the parameters over iterations, and draw the graph of the raw data and the line-of-best-fit. (4 marks)
- iii). Using your results in (i) and (ii), and predict the scores a student will obtain if he studies for 5 hours. (3 marks)

SECTION D

5. a). i). Distinguish between supervised and unsupervised learning. (2 marks)
- ii). Does more data necessarily mean more intelligence in machine learning? Briefly explain. (1 mark)
- iii). Differentiate between *batch learning* and *online learning* in machine learning. (2 marks)
- A lecturer wants to predict *Exam score* Y from *Study hours* X using the simple linear model. The following dataset is collected and shown in **Table 5a**. With initial parameters: $\theta_0 = 0$, and $\theta_1 = 0$, Learning rate $\alpha = 0.1$.
- b). i). Use the *Stochastic Gradient Descent* (BGD) to determine the parameters θ_0 , θ_1 fitting the SLR equation. Iterate over all the data in this order $A \rightarrow B \rightarrow C$. (4 marks)
- ii). Repeat (b). (i). using the order $A \rightarrow C \rightarrow D$. (4 marks)
- iii). Plot (b). (i) and (ii). on the same graph with the raw data. Predict the scores a student will obtain if he studies for 6 hours, and compare your results obtained from both line-of-best-fits. (2 marks)
- c). i). Differentiate between linear regression and logistic regression in terms of their output. (1 mark)
- ii). Given these three models; $p(h) = \beta_0 + \beta_1 h$, $\ln(p(h)) = \beta_0 + \beta_1 h$ and $\ln\left(\frac{p(h)}{1-p(h)}\right) = \beta_0 + \beta_1 h$. Discuss which of this model is used for logistics regression and why is such model used. (5 marks)
- iii). A university lecturer wants to model the relationship between *Study Hours* (X) and *Test Scores* (Y) using Simple Linear Regression (SLR). The dataset is provided in **Table 5b**. Based on the analysis, the following regression model was obtained: $\hat{y} = 49 + 4.5x$, where $\hat{y}$ is the predicted test score and x is the number of study hours. Using the data in **Table 5b**, compute the *coefficient of determination* R^2 for the regression model and interpret the significance of the R^2 value obtained in the context of this problem. (4 marks)
6. a). i). Discuss four real-life problems in Nigeria that can be solved using machine learning, and briefly state the ML approach suitable for each. (2 marks)
- ii). Briefly describe the *instance-based learning strategy* in machine learning, and give one example of a model that uses it. (2 marks)
- iii). In Supervised machine learning, differentiate between the terms *Target* and *Label*. (2 marks)
- A lecturer wants to predict *Exam score* Y from *Study hours* X with the SLR model. The lecturer collects this dataset shown in **Table 6a**. The parameters (θ_0 , θ_1) are to be estimated using *Mini-Batch Gradient Descent* (MBGD) with the following settings: *Batch size* = 2 (three batches); *Batch order*: $B1 = \{A, B\}$, $B2 = \{C, D\}$, $B3 = \{E, F\}$; *Learning rate* $\alpha = 0.1$; *Parameter Initialization* (θ_0 , θ_1) = (0, 0); *Number of epochs* = 1
- b). i). Perform one epoch of mini-batch gradient descent as specified above, updating the parameters after each batch. Present the updated values of (θ_0 , θ_1) at the end of the epoch. (6 marks)

- ii). Using the final parameters from (b.i): Plot the line of best fit together with the raw data points, and then predict the exam score of a student who studies for 8 hours. (2 marks)
- iii). Suppose the obtained parameters from (b.i) are adjusted such that θ_0 is increased by 0.2 and θ_1 is decreased by 0.2. Plot the new regression line together with both the original regression line from (b.i) and the raw data points, and predict the exam score for a student who studies for 8 hours using this adjusted model. (2 marks)
- c). i). State the role of the sigmoid function in logistic regression. (2 marks)
- A university lecturer wants to predict whether students will *Pass* (1) or *Fail* (0) a course based on their *hours of study* (X). A sample of students was observed as shown in **Table 6b**. Using *Maximum Likelihood Estimation* (MLE), the logistic regression model was fitted with the following parameters: $\theta_0 = -6$, and $\theta_1 = 1.2$
- ii). Plot the Sigmoid function for the obtained model parameters over the range of study hours relevant to **Table 6b**. Clearly indicate the decision boundary (probability threshold = 0.5). (5 marks)
- iii). Using the fitted model, compute the probability of passing for a student who studies 4 hours and for another who studies 8 hours. For each case, classify the student as *Pass* (1) or *Fail* (0) using a threshold of 0.5. (2 marks)

$$P(x) = \frac{1}{1 + e^{-x}}$$

$$x = \theta_0 + \theta_1 x$$

Table 1

Student	X1 (Visits)	X2 (%)	Y
A	1	55	Low
B	2	65	High
C	3	80	High
D	2	70	Low
E	4	85	High
F	5	95	High
G	3	60	Low
H	6	90	High
I	4	75	High
J	5	70	Low

Table 2

Student	X1 (Hours)	X2 (%)	Y
A	2	60	50
B	3	65	58
C	4	70	65
D	5	75	80
E	6	80	78
F	7	85	85
G	3	80	68
H	8	90	92
I	5	65	64
J	6	70	72

Table 3

Student	Hours (X)	Score (Y)
A	2	50
B	3	60
C	4	65
D	6	80
E	8	90
F	9	95

Table 4

Student	Hours (X)	Score (Y)
A	1	2
B	2	4
C	3	6

Table 5a

Student	X (Hours)	Y (Score)
A	1	3
B	2	5
C	3	7
D	4	9

Table 5b

Student	X (Hours)	Y (Score)
A	1	52
B	2	57
C	3	63
D	4	68
E	5	72

Table 6a

Student	X (Hours)	Y (Score)
A	1	3
B	2	5
C	3	7
D	4	9
E	5	11
F	6	13

Table 6b

Student	Hours Studied (X)	Outcome (Y)
A	2	0 (Fail)
B	3	0 (Fail)
C	4	1 (Pass)
D	5	1 (Pass)
E	6	1 (Pass)
F	7	1 (Pass)